

passing kernel derived from Linux, called the Little Kernel. This is the heart of Fuchsia and has been designed by Google to work with all kinds of devices like smartphones, laptops and IoT devices. This layer mediates hardware access, implements essential software abstractions over shared resources, and provides a platform for low-level software development.

Garnet: This is the second layer and provides device-level system services for software installation, administration, communication with remote systems, and product deployment. This layer contains the network, media and graphics services, and also contains the package management and update system.

Peridot: This third layer provides the services needed to create a cohesive, customisable, multi-device user experience assembled from modules, stories, agents, entities and other components. It contains the device, user, story runners and also the ledger and resolver, as well as the context and suggestion engines.

Topaz: This is the topmost fourth layer that increases system functionality by implementing interfaces defined by underlying layers. It contains four major categories of software: modules, agents, shells and runners. Modules include the calendar, email, and terminal modules; shells include the base shell and the user

shell; agents include the email and chat content providers; and runners include the Web, Dart, and Flutter runners.

Google Fuchsia is coded using the C, C++, Dart, Go, Rust and Python languages. It can support machines with ARM64 and x86-64 architectures. The Flutter development kit produces apps based on Dart that can run at 120 frames per second. Fuchsia also offers a Vulkan based graphics rendering engine called Escher, with specific support for 'volumetric soft shadows'. A special version of Android Runtime for Fuchsia will be developed, and it will run on machines with this system from a FAR file, which is the equivalent of the Android APK.

Security

Google Fuchsia uses a concept called sandboxing, which prevents the apps from gaining full access to the OS. All access to the kernel of the OS is exposed to the apps as object-capabilities, which means that applications running on Fuchsia have no ambient authority — they can interact only with the objects to which they have been granted access explicitly. Software is delivered in hermetic packages and everything is sandboxed, which means all software that runs on the system, including applications and system components, gains access only to the information it needs to know. Such architecture keeps

the kernel safe, and can prevent any venerable attacks.

Features of Fuchsia

- Fuchsia's user interface and apps are written with a software development kit called Flutter, and it allows cross-platform development abilities for Fuchsia, Android and iOS.
- Due to the cross-platform opportunities offered by the Flutter development kit, users are able to install parts of Fuchsia on Android devices also.
- Fuchsia packages are designed to be updated independently or even delivered ephemerally, and the software is always up-to-date, like a Web page.
- Fuchsia currently supports many types of languages and runtimes, including C++, Rust, Flutter and Web. Developers can use a variety of languages or runtimes without needing to change Fuchsia itself.
- Fuchsia is designed for optimised performance. The use of asynchronous message passing communication reduces latency by letting the sender proceed without waiting for the receiver. It also optimises memory use by avoiding garbage collection in the core operating system, which helps to minimise memory requirements to achieve performance equivalent to other OSs.

Google Fuchsia is a new open source RTOS that will come with various improved features compared to Android, and can help various IoT devices to work effectively. This operating system was chosen by Google to overcome the drawbacks of the Android OS, and that's why it will give us a better user experience. **END** 

Pros and cons of Fuchsia

Pros	Cons
• Fuchsia supports many languages like C, C++, Dart, Go, Rust and Python. • It is developed using a development kit called Flutter, which can also be used to create cross-platform apps for Fuchsia, Android and iOS. Fuchsia can also support Android apps (APK) by using a runtime. • It can support both ARM-46 and x86-64 architecture and can run on laptops, smartphones and IoT devices. • Fuchsia uses a sandboxing concept to run the apps, and this makes it a secure OS by preventing any attacks.	• Fuchsia uses a message passing kernel instead of shared memory, the system has slower communication and the connection time between the processes may take some time. • Fuchsia uses a UI that is different from Android, and current Android users might find it difficult to use.

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